UC San Diego
RADY SCHOOL OF MANAGEMENT

MGTF 423 Data Science for Finance I

Winter 2026

Amarnath Gupta
A1gupta@ucsd.edu

On zoom offered through Canvas

TA: Yifan Mao : Y3mao@ucsd.edu
Bhaavya Naharas :bnaharas@ucsd.edu

DESCRIPTION

Data Science for Finance – I is a graduate-level course designed for MBA Finance students that equips them with practical skills in managing and analyzing both structured and unstructured data for modern financial applications. The primary goal of the course is to develop proficiency in data management using SQL and in algorithmic text analytics applied to financial documents such as prospectuses, annual reports, regulatory filings, and contracts. As data science and AI increasingly shape finance and FinTech roles, this course addresses a critical gap in the traditional business school curriculum, where systematic handling of large-scale structured and textual data is often underemphasized. The course is strongly applied and practitioner-oriented, reflecting industry expectations for hands-on competence with real data and tools. Instruction combines lectures focused on SQL-based data manipulation with a substantial group project centered on text analytics, where students programmatically use large language models to analyze real-world financial documents and extract insights relevant to contemporary financial decision-making.

OBJECTIVES

At the close of this course, you will be able to:

- Query database systems using SQL
- Use Python to connect with relational database systems
- Perform analytical operations with SQL
- Perform text analytics using Python libraries
- Use LLMs and related tools to answer natural language questions on text
-

MATERIALS

Required

- All required reading materials will be supplied in class
- An API key for an LLM like OpenAI will be needed for each project group

Recommended

- SQL for Data Analytics: Harness the power of SQL to extract insights from data, Third Edition
  By Jun Shan, Matt Goldwasser, Upom Malik, Benjamin Johnston

UC San Diego
RADY SCHOOL OF MANAGEMENT

SCHEDULE

Date	Due	Class Topic & Activities
Week 1 (Jan 6)		Introduction to Data Models and the Relational Data Model
Week 2 (Jan 13)		Introduction to SQL
Week 3 (Jan 20)	Homework 1	Advanced SQL – 1
Week 4 (Jan 27)	Homework 2	Advanced SQL – 2
Week 5 (Feb 3)	Homework 3	Analytics with SQL
Week 6 (Feb 10)		Python Refresher
Week 7 (Feb 17)	Mid-Term Exam due Feb 16th	Introduction to Text Analytics
Week 8 (Feb 24)		Statistical Methods in Text Analytics
Week 9 (Mar 3)		NLP Methods in Text Analytics
Week 10 (Mar 10)		Project Discussion
Finals (Mar 17)		Final Project Presentation

ASSIGNMENTS

Programming Assignments
All programming assignments are individual and will be submitted through Canvas.

Team Projects
The Final project is a group project ideally with three students per team.

Exams
Midterm exam will be taken home offered through Canvas. You will have 24 hours to complete and submit on Canvas. Academic Integrity will be strictly enforced. Midterm will cover the SQL and Data Management part of the course.

Class Participation Attendance and Participation
Students are expected to attend the class in person. While absence from the class will not be penalized, the students who are absent may miss out on classroom discussions.

GRADING

Assignments	Percentage %
Program Assignment 1	15%
Program Assignment 2	15%
Program Assignment 3	15%
Midterm Exam	15%
Final Project	40%
Total	**100%**

CHATGPT AND OTHER AI TOOLS

Large Language Models and AI in general are new tools that can answer simple questions about economics and economic problems, however, they can make simple mistakes too. Since their growth is recent (Fall 2022 and early 2023), we are developing an AI policy, mainly thinking about ChatGPT.

If you find it helpful, you are welcome to try to use ChatGPT (or similar tools) to offer advice or additional support on your homework and cases. However, you must ultimately do the work on your own. For group homework, you are expected to do the write-up of the case on your own. The TA will check your writing and raise flags if it is a copy of chatGPT/AI output.

The midterm and final exams will emphasize your ability to understand and implement economic analysis tools for managerial problems. You are expected to work on your own, without chatGPT/AI for the SQL part of the class. You are expected to use OpenAI and related software for class projects. **SQL for Data Analytics: Harness the power of SQL to extract insights from data , Third Edition**

Jun Shan Matt Goldwasser Upom Malik Benjamin Johnston

COURSE POLICIES

We expect students to attend classes in person.

We will have 4 classwork assignments where students will be required to complete an assignment in class. The dates will be announced in class in advance.

A homework assignment can be late by at most one day. After that we will reduce the score by 10 points for each additional day missed.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

UC San Diego
RADY SCHOOL OF MANAGEMENT

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or **osd@ucsd.edu.**